UV-C Vessel Propulsion System

https://github.com/comp195/Spring2020Project-alex-emiliano/wiki

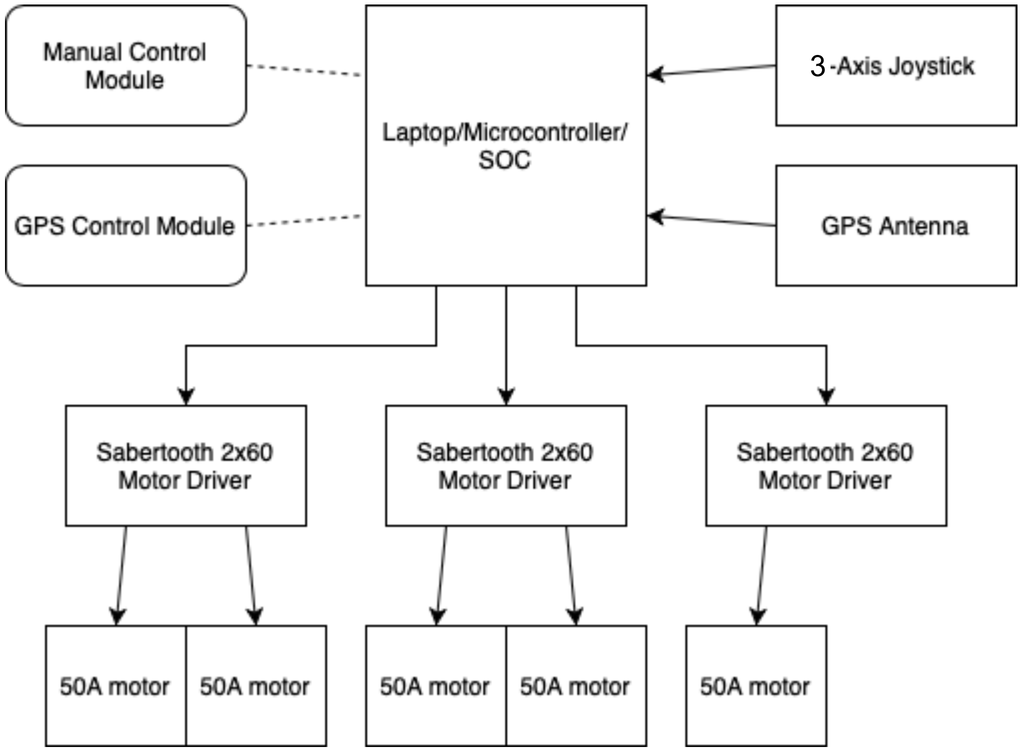
Alexander Ng - a_ng9@u.pacific.edu
Emiliano Barroso - e_barroso@u.pacific.edu

Last Updated: Apr. 28, 2020

Introduction

The Tahoe Resource Conservation District leads aquatic invasive plant control efforts in the Lake Tahoe Basin. Current techniques in curtailing growth of underwater techniques such as diver-assisted suction are both extremely costly as well as time consuming. Inventive Resources Inc. (IRI) has proposed an alternative method using UV-C light to kill invasive plants. By mounting an array of UV-C lights under a mobile floating platform, they are able to treat larger areas more efficiently. This project streamlines and improves workflow for the vessel by designing a manual control system to efficiently control all eight motors with a single joystick, as well as incorporating GNSS-based navigation to automate the vessel.

System Architecture



Manual Control Module: Arduino program to facilitate user control of motors

GPS Control Module: Electron application to automate control of motors using GPS and display output to screen

Laptop/Microcontroller: Windows laptop connected to Arduino Mega - electron code is run on laptop and input/output read from and to Arduino pins during GPS mode, and firmware is written to Arduino during Manual mode.

3-axis joystick: input control for manual control module - the one joystick will control an array of motors. X, Y, and Z axes used for translational and rotational movement

GPS antenna: collects GPS data in NMEA format for automation. GPS module used is ublox's C099-F9P board.

Motor Drivers: Sabertooth 2X60 motor drivers read commands from arduino and power motors appropriately. Each motor driver needs to be connected to an external power supply and can control two motors each. Four motor drivers are used in total to control an array of eight motors.

Hardware, Software, System Requirements

- Hardware requirements: Laptop, GPS module, Arduino Mega, Motor Array, Joystick
- System requirements: Windows 10

Ideally, this system will be able to be platform-agnostic and low-spec - any PC that can connect to the Arduino will be able to run the driver program.

External Interfaces

3-Axis Joystick: Gives three analog signals for X/Y/Z. Will be read by analog pins on Arduino

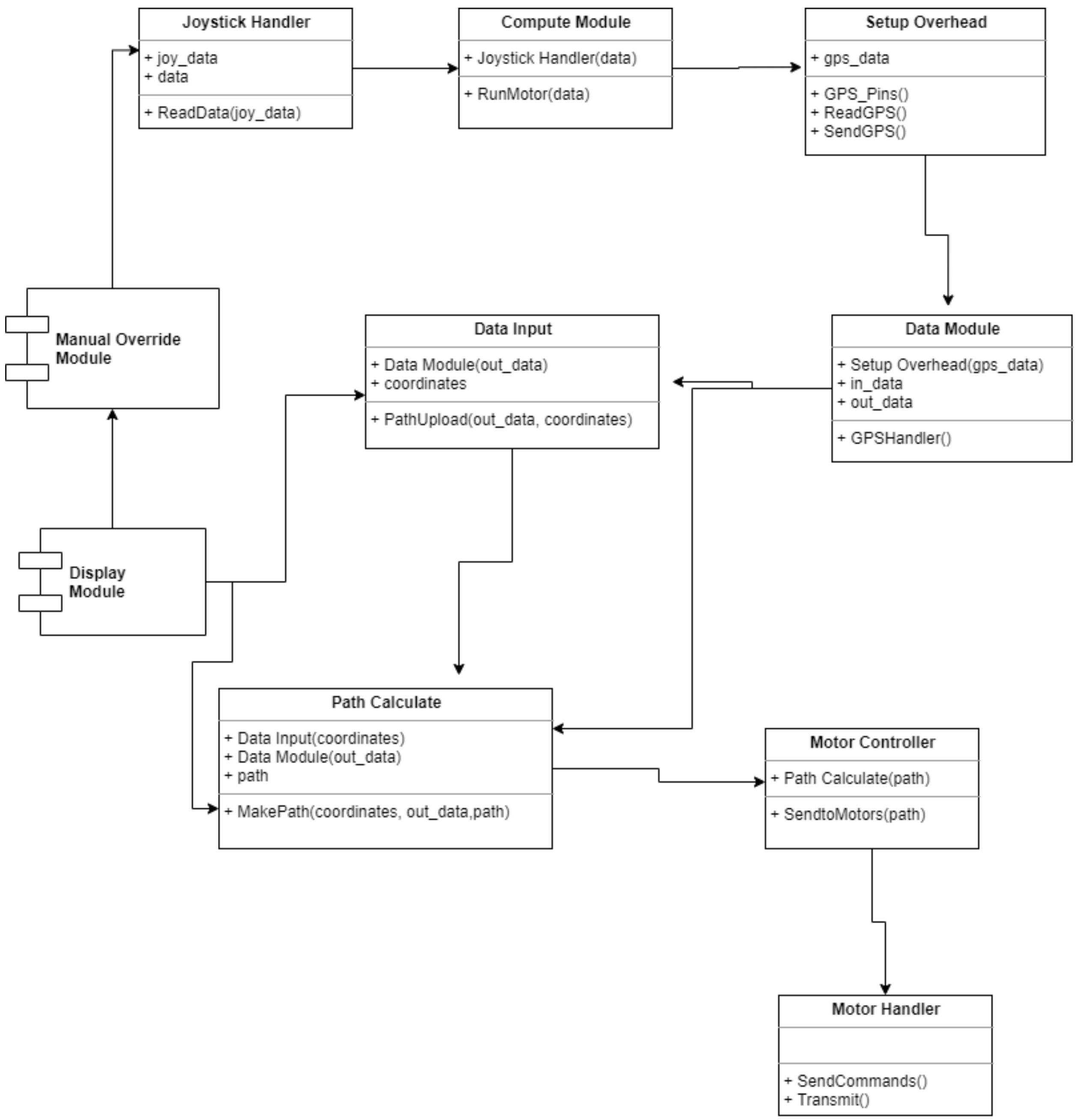
Sabertooth 2x60 motor drivers: Motor drivers must be wired with 8 ga. wire to avoid heat buildup. Drivers will be connected to TX pin on Arduino and packetized serial will be used to communicate with individual motors. See References for datasheet.

C099-F9P GNSS board: Data will be read in by GPIO pins on Arduino in NMEA format. NMEA format contains GPS and time data. See References for datasheet

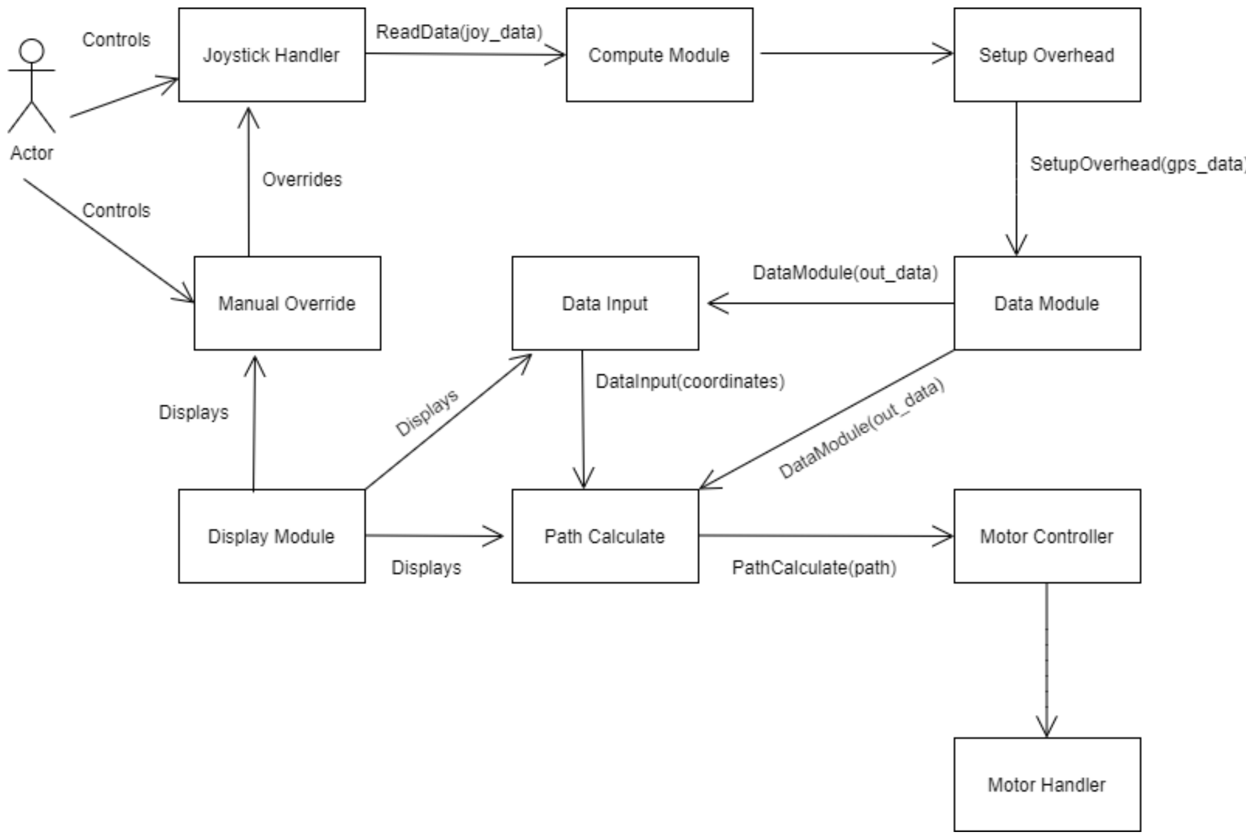
Arduino Mega: Microcontroller used to send commands to pins and read joystick/GPS input. See References for technical specs.

Software Design

Class Diagram



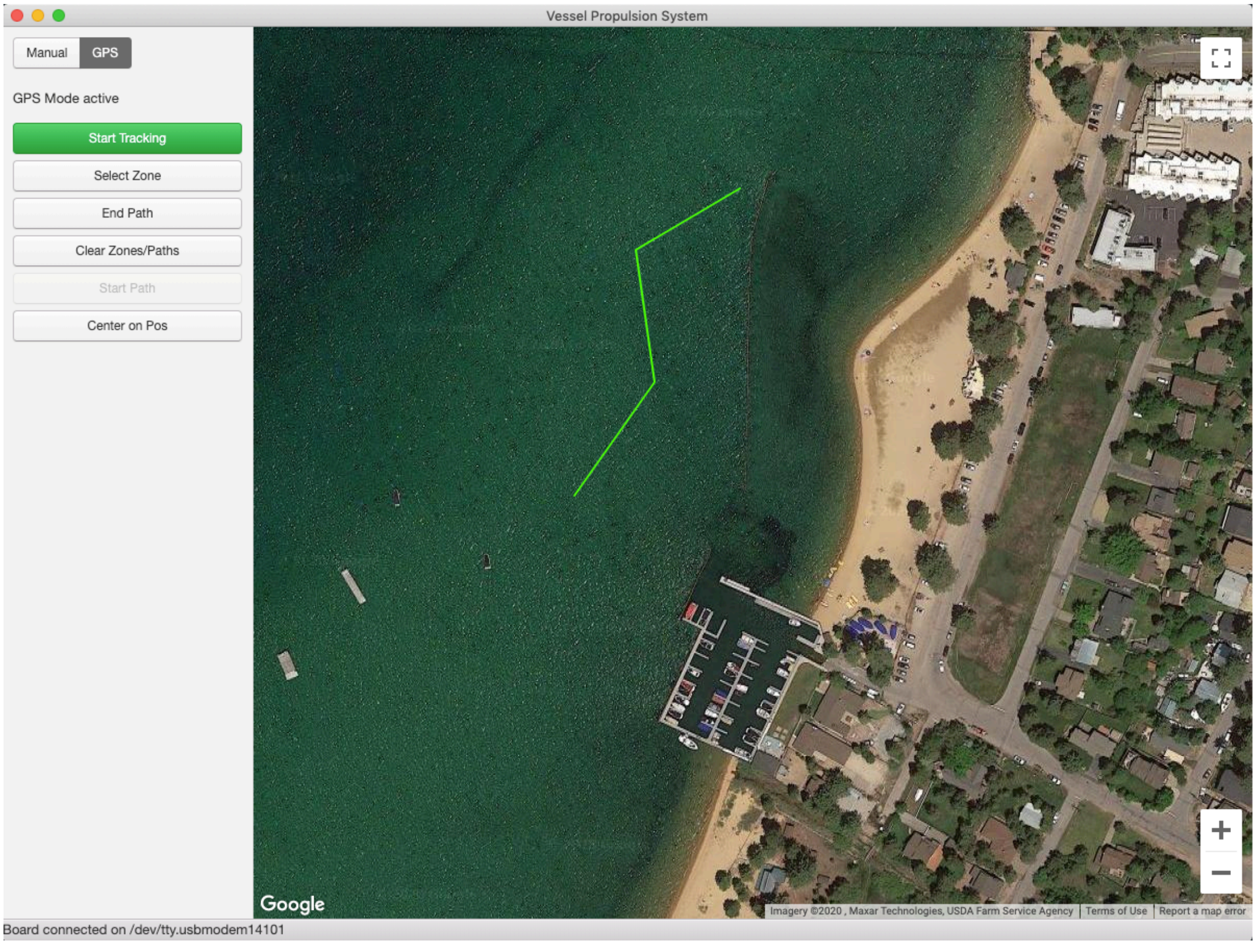
Interaction Diagram



Design Considerations

There are three individual components that make up the project: Headless-Manual, Arduino-Manual, and GPS modes. Headless-Manual utilizes the motor driver's onboard logic to drive the motor array and thus requires no code to be written, but determines the physical wiring of the motor drivers. This mode can only be controlled manually. Arduino-Manual utilizes programs written to the Arduino to parse joystick data. This mode is also only controlled manually. GPS mode uses an Electron application as a GUI that parses data from both GNSS board as well as joystick – this mode supports both manual control and GNSS automation.

User Interface Design



The GPS module will feature a simple user interface with a column of buttons for individual commands – it is accompanied by a visual representation of the floating vessel and the path it is following in real time overlaid on a map of the terrain.

Manual control module will not have a user interface (apart from console debug) as the user will control the boat visually with the joystick.

Glossary of Terms

Arduino Mega: A microcontroller board that enables users to create interactive electronic objects.

GPIO Pins: General Purpose Input Output, a customizable circuit that can be controlled by software.

NMEA: National Marine Electronics Association. Used to refer to NMEA 0183, a standard output protocol for GPS data. The GPS antenna used in this project will output data in this format.

GNSS: Global Navigation Satellite Systems – the catch-all name for GPS-type navigation systems (GPS is a US-based system, but other countries have their own: GLONASS/Russia, BDS/China, Gallileo/EU). The board in use is capable of receiving data from all satellite constellations

References

Ada, L. (n.d.). Adafruit Ultimate GPS. Retrieved from https://learn.adafruit.com/adafruit-ultimate-gps/overview

Arduino. (n.d.). Arduino Mega 2560 Rev3. Retrieved from https://store.arduino.cc/usa/mega-2560-r3

Dimension Engineering. (2011). Sabertooth 2X60 User's Guide. Retrieved from https://www.dimensionengineering.com/datasheets/Sabertooth2x60.pdf

ublox. (2019). C099-F9P User Guide. Retrieved from https://www.u-blox.com/sites/default/files/C099-F9P-AppBoard-Mbed-OS3-FW_UserGuide_%28UBX-18063024%29.pdf